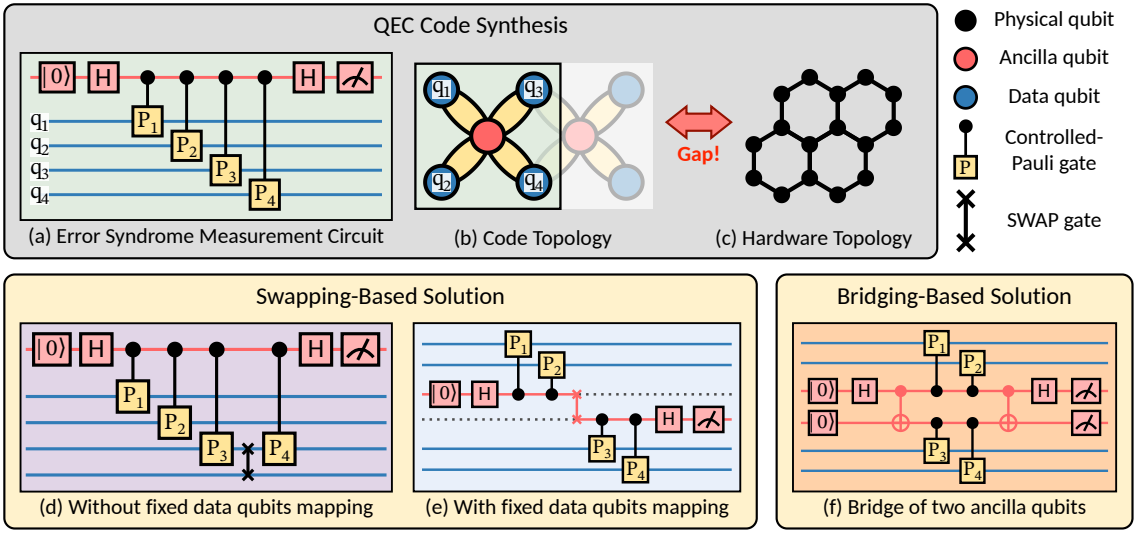

QEC Code Synthesis
|0⟩
H
P1
P2
P3
P4
H
q1
q2
q3
q4
(a) Error Syndrome Measurement Circuit
q1
q3
q2
q4
(b) Code Topology
Gap!
(c) Hardware Topology
Physical qubit
Ancilla qubit
Data qubit
Controlled-Pauli gate
P
SWAP gate
Swapping-Based Solution
(d) Without fixed data qubits mapping
(e) With fixed data qubits mapping
Bridging-Based Solution
(f) Bridge of two ancilla qubits